



Guideline

Methotrexate Monitoring for Inflammatory Bowel Disease Patients

Scope

Site	Service/Department/Unit	Disciplines
SCGOPHCG	Gastroenterology	Medical, Nursing, Pharmacy

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Definitions

Inflammatory Bowel Disease (IBD)	Characterised by chronic inflammation of the gastrointestinal tract. Comprised of Crohn's disease (CD), ulcerative colitis (UC) and inflammatory bowel disease – unclassified (IBD-U).
Crohn's Disease (CD)	Type of IBD causing inflammation to the gastrointestinal tract, more commonly in the small intestine.
Ulcerative Colitis (UC)	Type of IBD causing inflammation and ulcers to the innermost lining of the large intestine.

Summary

- Methotrexate is used as immunomodulatory agents to exert a steroid sparing effect and to prevent immunogenicity to biological therapies in patients with IBD.
- Methotrexate is dosed **once weekly**.
- Blood tests should be monitored at weeks 1,2,3,4,6,8,12 during initiation, then if dose and bloods are stable patients can be discharged to the care of GP for 3 monthly blood tests thereafter.

Week 1	FBP, LFT U&Es and CRP
Week 2	FBP, LFT U&Es and CRP
Week 3	FBP, LFT U&Es and CRP
Week 4	FBP, LFT U&Es and CRP
Week 6	FBP, LFT U&Es and CRP
Week 8	FBP, LFT U&Es and CRP
Week 12	FBP, LFT U&Es and CRP

1.0 Purpose

A guideline for pharmacist led methotrexate monitoring in Inflammatory Bowel Disease (IBD) patients.

To maintain remission in patients with Inflammatory Bowel Disease (IBD), comprising of Crohn's disease (CD), ulcerative colitis (UC) and inflammatory bowel disease-unclassified (IBD-U).

Methotrexate is also used to augment anti-TNF therapy, in part by reducing immunogenicity of biologic agents.

2.0 Presentation

Tablets: 2.5mg, 10mg (scored)

Subcutaneous injections: 7.5mg/0.15mL, 10mg/ 0.2mL, 15mg/0.3mL, 20mg/0.4 mL, 25mg/0.5mL. (pre-filled syringes)

Subcutaneous injections are the preferred method of administration for WEEKLY doses >15mg, due to greater bioavailability when compared to oral tablets¹.

NOTE: Pre-filled syringes are non-PBS in this cohort of patients and must be prescribed on a hospital prescription and dispensed by the pharmacy department.

3.0 Administration

Methotrexate is given once a week on the same day of each week.

All patients commencing methotrexate will be provided with education by the IBD pharmacist or IBD nurse prior to commencement.

The IBD nurse/pharmacist will educate the patients on subcutaneous injection techniques. Patients will be supplied the Trexject® consumer Medicines Information Leaflet (CMI) which outlines step-by-step instructions for self-administration. Patients with concerns about self-administering subcutaneous methotrexate are encouraged to attend their GP for administration aid.

Doses $\leq$ 15mg weekly can be prescribed as an oral or subcutaneous dose.

Doses $>$ 15mg to 25mg weekly are best tolerated and absorbed when given subcutaneously.

Folic acid supplements are prescribed to reduce the gastrointestinal side effects and to reduce the liver toxicity from methotrexate.

Folic acid: minimum of 5mg to 10mg weekly (not taken on the same day as methotrexate).

(Commonly prescribed in higher doses: e.g. folic acid 5mg daily except methotrexate days).

4.0 Pre-Treatment Screening

All IBD patients commencing immunomodulator therapy require baseline pre-screening. This is particularly useful if patients require rapid escalation to a biologic therapy.

All results should be reviewed by a consultant gastroenterologist.

Baseline Bloods	• FBE • LFTs • CRP
Infection Serology	• Hepatitis A, B, C • HIV • Varicella Zoster • EBV • CMV • TB screening (chest X-ray and QuantiFERON-Gold)
Check Vaccination Status is up to date	• Pneumococcal vaccine • Influenza • Human Papilloma Virus (in >50yrs) • VZV Varicella vaccine if negative serology or no history of vaccination, shingles or chickenpox. • Hep B in seronegative patients • Tetanus (if not administered in last 10 years) NOTE: Live vaccines such as MMR, typhoid, yellow fever, BCG and varicella (Zostavax) should be avoided during thiopurine therapy. Inactive/attenuated vaccines are safe for administration during treatment but may need booster doses. Consult the local immunisation control program office if any advice is needed.

5.0 Risk Factors for Methotrexate therapy

Assess the risk factors for methotrexate therapy:

Liver disease	Increased association with hepatotoxicity in the context of NAFLD and risk of pre-existing liver fibrosis.
Excess alcohol intake	Alcohol increases the risk of liver damage while taking methotrexate. Methotrexate usage in heavy drinkers has been associated with cirrhosis of the liver.
Obesity, diabetes, high fasting blood glucose, lipid profile	Risk factors for the development of NAFLD and methotrexate induced liver toxicities.
Women of childbearing age	Methotrexate can cause foetal abnormalities and should not be taken by women who wish to conceive. Adequate contraception must be used in women of child-bearing age.

Males planning conception	Can cause reversible oligospermia but the clinical relevance is not clear. Consider ceasing 3 months prior to conception in males with well-controlled IBD after discussion with consultant gastroenterologist.
FBP, U&Es, LFTs and Albumin	Methotrexate can cause immunosuppression. Baseline bloods are required prior to commencing therapy. Methotrexate undergoes renal clearance and there is greater risk of toxicity in patients with reduced renal function. Methotrexate is associated with thrombocytopenia, liver and pulmonary toxicities.
Chest X-Ray	After discussion with a consultant gastroenterologist, obtained within the previous year to rule out interstitial lung disease in patients considered to be at risk (based on past medical history).

6.0 Contraindications

Avoid use in patients with known liver disease (including fatty liver) and known excess alcohol use.

Females planning pregnancy or who are breast-feeding.

Active and recurrent infections (bacterial, viral and fungal).

Pre-existing myelosuppression.

7.0 Blood Monitoring Parameters

Event (Occurrence)	Recommended Action
Total WCC $<3 \times 10^9/L$ OR Neutrophil count $<1.5 \times 10^9/L$	Ask patient about any signs of infection, discuss with consultant gastroenterologist. Repeat FBP in 1 week. If stable repeat in one month, and then return to 3-monthly monitoring as usual. In cases of profound myelosuppression (e.g. WCC $<2.5 \times 10^9/L$ and/or neutrophil count $<1 \times 10^9/L$), cease medication and discuss with consultant gastroenterologist on the same day.
Lymphocyte count $<0.5 \times 10^9/L$	Immunomodulator induced severe lymphopenia ($<0.5 \times 10^9/L$) is a positive indicator for severe infections. Monitor for additional signs of myelosuppression or infections and consider dose reduction or cessation, depending on clinical picture. To be discussed with consultant gastroenterologist.
Platelet count $<100 \times 10^9/L$	Low platelet count may be a consequence of widespread myelosuppression. Withhold

	treatment and discuss with consultant gastroenterologist.
MCV >105 fL	Check serum vitamin B12, folate and TFT and discuss with consultant gastroenterologist.
Abnormal LFT	Patients should be screened for pre-existing liver disease prior to commencing methotrexate. Avoid use in patients with known liver disease. Methotrexate induced hepatotoxicity can be mild and self-limiting with or without dose adjustments but may develop into fatty liver, fibrosis or cirrhosis. If LFTs are < 2x the upper limit of normal (ULN), do not stop or change dose, recheck LFTs in one to two weeks. If LFTs are > 2x ULN, contact consultant gastroenterologist and consider interrupting/ceasing therapy. Consider liver biopsy and hepatology review.
Hypoalbuminemia (unexplained fall in the absence of active disease)	Monitor closely and consider need for liver biopsy.

8.0 SHIMS Database

Enter patient weight and height results into SHIMS database.

Ensure all pre-treatment screening are completed and enter results into SHIMS database.

Automated reminders from SHIMS for bloods and blood monitoring protocol are outlined below. The first reminder is sent out to the patient the following Monday after initial commencement.

Blood monitoring via SHIMS to be completed weekly for the first 4 weeks, then fortnightly for a month, then at week 12 followed by three-monthly thereafter.

Restart monitoring schedule after any dose escalation.

Week 1	FBP, LFT U&Es and CRP
Week 2	FBP, LFT U&Es and CRP
Week 3	FBP, LFT U&Es and CRP
Week 4	FBP, LFT U&Es and CRP
Week 6	FBP, LFT U&Es and CRP
Week 8	FBP, LFT U&Es and CRP
Week 12	FBP, LFT U&Es and CRP

Monitor for adverse effects as outlined below under “9.0: Adverse Reactions and Recommended Actions”.

Once a patient has completed the initial 12 weeks of blood monitoring and the dose and blood results are stable, then the patient can be discharged to the care of their GP for ongoing three-monthly monitoring of bloods (FBP, LFT, U&Es and CRP).

9.0 Adverse Drug Reactions and Recommended Actions

Event (Occurrence)	Recommended Action
Total WCC $<3 \times 10^9/L$ OR Neutrophil count $<1.5 \times 10^9/L$	Ask patient about any signs of infection, discuss with consultant gastroenterologist. Repeat FBP in 1 week. If stable return to monitoring as per guideline. In cases of profound myelosuppression (e.g. WCC $< 2.5 \times 10^9/L$ and/or neutrophil count $<1 \times 10^9/L$, cease medication and discuss with consultant gastroenterologist on the same day.
Lymphocyte count $<0.5 \times 10^9/L$	Immunomodulator induced severe lymphopenia ($<0.5 \times 10^9/L$) is a positive indicator for severe infections. Monitor for additional signs of myelosuppression or infections and consider dose reduction or cessation, depending on clinical picture. To be discussed with consultant gastroenterologist.
Platelet count $<100 \times 10^9/L$	Low platelet count may be a consequence of widespread myelosuppression. Withhold treatment and discuss with consultant gastroenterologist.
MCV >105 fL	Check serum vitamin B12, folate and TFT and discuss with consultant gastroenterologist.
Abnormal LFT	Patients should be screened for pre-existing liver disease prior to commencing methotrexate. Avoid use in patients with known liver disease. Methotrexate induced hepatotoxicity can be mild and self-limiting with or without dose adjustments but may develop into fatty liver, fibrosis or cirrhosis. If LFTs are $< 2x$ the upper limit of normal (ULN), do not stop or change dose, recheck LFTs in one to two weeks. If LFTs are $> 2x$ ULN, contact consultant gastroenterologist and consider interrupting/ceasing therapy. Consider liver biopsy and hepatology review.
Concurrent infection	In cases of acute bacterial or viral infection consider withholding methotrexate for a short period. If infections are severe and requiring hospital admission and intravenous antibiotics withhold methotrexate immediately. Discuss with consultant gastroenterologist.

Nausea/Vomiting	Commonly occurs. Mild symptoms will often recede with time. Consider taking methotrexate with food or in the evening before bed or splitting the dose of methotrexate to twice a day on the day of the week the methotrexate is due. The subcutaneous injections are better tolerated than tablets. Ensure adequate intake of folic acid supplementation. If severe consider stopping medication and discussing with consultant gastroenterologist.
Skin rashes	Assess rash for herpes zoster (shingles). If shingles is suspected, withhold methotrexate and refer to GP within 72 hours to start a course of antiviral.
Severe sore throat and abnormal bruising	Repeat FBP to rule out bone marrow suppression.
Mouth ulcers	The use of folic acid supplements makes this occurrence less likely. Assess the severity and withhold dose if appropriate and reassess in 1 week.
Renal impairment – significant deterioration compared to baseline or ULN	Withhold treatment and reassess immediately, Discuss with consultant gastroenterologist.
New or increasing dyspnoea or dry cough in the clinical context of interstitial fibrosis concern	Further clinical assessment required. Withhold treatment as necessary, discuss with consultant gastroenterologist and consider arranging chest x-ray and pulmonary function test.
Mild tiredness, headache and mental clouding and or muscle/joint pain after the weekly dose	This is usually temporary after the dose and improves with time. It may worsen temporarily if the dose is increased. If there is no improvement, the dose may need to be reduced.

10.0 Important Drug Interactions with Methotrexate

Medication	Interaction	Recommendation
Antibiotics such as Trimethoprim or Sulfamethoxazole with Trimethoprim	Can increase the toxicity and side effects of methotrexate.	Recommend another antibiotic if possible.
Probenecid	Probenecid inhibits renal tubular secretion of methotrexate and may increase its serum concentration and therefore toxicity and side effects. Weekly doses of methotrexate pose a lower risk of toxicity than antineoplastic daily dosing ⁵ .	Consider alternatives to probenecid if possible. If given concurrently, consider dose reduction of methotrexate and monitor for side effects.
Sulfasalazine	Risk of hepatotoxicity increased.	Ensure adequate folic acid replacement (10mg weekly) and monitor for hepatotoxicity.
Alcohol	Alcohol increases the risk of liver damage while taking methotrexate. Methotrexate usage in heavy drinkers has been associated with cirrhosis of the liver. It is not known precisely what level of drinking is safe when on methotrexate, however there is general agreement that 1-2 standard drinks taken once or twice a week is unlikely to cause a problem. Drinking more than 4 standard drinks on one occasion, even if infrequently is strongly discouraged.	
Live Vaccines	Live vaccines should not be co-administered during methotrexate therapy as there is an increased risk of infection and developing adverse events in immunocompromised	Decisions regarding live vaccines should always be made in consultation with a gastroenterologist.

	patients. Vaccination status should be up to date prior to commencing methotrexate.	
NSAIDS	Methotrexate has its toxicity increased by NSAIDS.	Use combination with extreme caution. Monitor for signs of methotrexate toxicity.

11.0 Use in Pregnancy

Methotrexate must not be taken during pregnancy as it can cause miscarriage or foetal deformity.

Women of child bearing age should use effective contraception while taking methotrexate. Women planning to become pregnant should stop taking methotrexate 3 months before attempting to conceive.

Methotrexate does not affect a person's ability to have children in the long term.

Methotrexate produces oligospermia in males that will improve within a few months after stopping the methotrexate.

It is recommended males cease methotrexate 3 months prior to conception but decisions should be based on clinical picture and discussed with the consultant gastroenterologist.

12.0 Use in Breastfeeding

Methotrexate is not recommended during breastfeeding as it is excreted into the breast milk and accumulates in the neonate.

13.0 Storage and Handling

Keep out of reach of children and store below 25°C.

Methotrexate is not film coated. Health professionals handling methotrexate for patient administration should not divide, crush or break the tablet.

Patients must be educated on the disposal requirements of methotrexate syringes in a sharps container.

Related Documents

Legislation and policy:

- [Medicines and Poisons Act 2014](#)
- [Medicines and Poisons Regulations 2016](#)
- [Health Services Act 2016](#)
- [SCGH Hospital Policy 141 Medicines Management](#)
- [Mandatory Policy MP0077/18 Statewide Medicines Formulary Policy](#)

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